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Concurrent Nonparametric Estimation of Organ Geometry and Tissue Stiffness Using Continuous Adaptive Palpation

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Abstract— **Surgeons often manually palpate tissue or organs in order to find tumors or other anatomical structures. Information about organ geometry and tissue stiffness gained from palpation can also be extremely useful in robotic surgery for diagnosis, surgical guidance, and registration to other preoperative information. However, it is not always easy to obtain, even if the robot is equipped with force sensors. This paper reports our approach for concurrent estimation of stiffness and surface geometry, using a continuous motion similar to a sweeping palpation motion used by surgeons. Our method relies on force data captured by a tactile sensor rigidly attached to an end-effector probe. We use Gaussian processes to simultaneously estimate geometry and stiffness. The method is not tied to any specific robotic platform and is consistent with a variety of palpation strategies. For simplicity, we discuss the results based on two different palpation primitives. This is our first step towards developing an adaptive high fidelity model reconstruction and path optimization technique.**

I. INTRODUCTION

Estimating the stiffness of tissues is extremely useful for surgeons wishing to identify tumors or other anatomic structures. In traditional open surgery, surgeons palpate tissue with their fingers, typically with a sweeping or circular motion [1]. In minimally invasive or robotic surgery, this direct access to tissue and haptic feedback is lost, making it much harder to characterize tissue properties [2]. A number of authors (e.g. [3]–[7]) have developed finger-like tactile and force sensors to provide information about tool-to-tissue interaction forces. Tissue stiffness at a point is determined by measuring the change in force as the probe is pressed into the tissue. Mavash et al. [8] developed a control system for da Vinci surgical system [9], which provides force feedback with a position-position controller with friction and inertia compensation. Using this system they provided some results on stiffness estimation of a tissue model, based on discrete palpation. More recently Xu and Simaan have developed methods for estimating the force/moment acting at the tip of continuum robots by using measurements of joint-level actuation forces and extrinsic information regarding the type of interaction with the environment [10], [11]. These methods have been in used [12] to enable force-controlled exploration of flexible anatomy in a manner that benefits from the palpation methods presented in this paper.

Generally, the surface geometry of organs being manipulated is also important to support registration of preoperative image data, "augmented reality" information support to the surgeon, and improved robot control. Many authors (e.g., [13]–[15]) have reported computer vision methods for estimating surface geometry. Additional information may be obtained simply from the robot's end-effector position at the time tissue is contacted, and this information may be used to correct for camera-to-robot misalignment. By probing tissue at a large number of discrete points, it is also possible to build up a model of tissue stiffness across an organ. However, this can be extremely time consuming and wasteful.

Gaussian Processes (GPs) are a standard statistical learning technique for probabilistic modeling where the model parameters can be inferred directly from the data [16]. GP models have proven effective for nonparametric regression, classification, and other complex statistical learning tasks. A key advantage of the GP approach lies in the ability to provide informative confidence intervals on predicted values thus quantifying the quality of reported estimates. In this work we employ GPs in two alternative ways, termed *direct* and *latent*: 1) by directly learning a force function over the 3D tissue domain using force measurements collected along palpation paths, and then inferring geometry and stiffness from the resulting force model, and 2) by constructing two *latent* GP models over the unknown geometry and stiffness and using a force-stiffness physical relationship to optimally and concurrently estimate the hidden GP parameters using force measurements. We then propose an adaptive tissue sampling strategy by performing path optimization guiding the palpation to the most informative regions.

In complementary work to that reported here, our collaborators at Carnegie Mellon University are exploring the use of GP in a Bayesian optimization framework to identify stiff regions by probing the organ at a smaller number of discrete locations while simultaneously registering the end effector to an a-priori geometric model of the organ [17]. One advantage of discrete probing is that it permits accurate assessment of tissue stiffness and surface location at the point probed. The drawback is that it can be inefficient, compared to continuous palpation.

Although continuous palpation can easily yield dense force information along the palpation path, concurrent estimation of tissue stiffness and surface geometry becomes more difficult, since each measured force corresponds to a different

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surface position, and estimation away from the path also needs to be considered. In this paper, we report the use of GPs to provide a concurrent probabilistic estimate of surface geometry and tissue stiffness, based on continuous palpation with a force measurement probe. The palpation motion causes the probe to move laterally across the tissue while varying the depth of surface indentation, which can be done by varying the force exerted by the probe on the tissue or by a variety of position control strategies. Tissue stiffness and geometry, together with corresponding uncertainties, are readily computed from the force field estimate.

II. METHODOLOGY

In this section we list the proposed model assumptions, describe the robot setup, and provide details for the two palpation primitives used to collect data for validation purposes.

A. Model Assumptions

Our goal is to construct predictive probabilistic models of surface geometry and stiffness, by estimating the force field, through palpation. For our experiments we used a phantom(Fig 1) made of silicon elastomer (M-F Manufacturing, Fort Worth, Texas). In this initial development we made the following simplifying assumptions: 1) stiffness can be computed only using the magnitude of the force along a normal direction and depth of contact point, and 2) the relationship between force and stiffness can be well approximated by a linear model of the form

$$f_2 - f_1 = \kappa(d_2 - d_1), \tag{1}$$

where f_1 and f_2 are the forces at a depth d_1 and d_2 from the tissue surface, respectively and κ is the stiffness value at a surface point s.

B. Setup

This work is intended to be applied to any robotic platform, either by attaching a force sensor probe to the end-effector or by using other methods (eg [12]) to estimate tool-tissue interaction force. Our implementation employs a simple cartesian robot, with 3 linear stages, as shown in Fig 1. An ATI Mini-40 Force Torque sensor is attached at the end-effector for sensory information. The 3 joints of the robot are actuated using Galil Motion Controller and the end-effector probe is driven using admittance control. A simple form of admittance control is given by

$$\delta v_f = \mathcal{G}(F_c - f), \tag{2}$$

a where δv_f is the incremental velocity calculated based on contact force F_c detected by the force sensor and and some constant desired/limiting force f. $\mathcal{G}$ is the gain matrix which can be manipulated to change the velocity of the robot.

C. Palpation

Palpation is achieved using motion primitives corresponding to sweeping motions at variable depths, e.g. following a sinusoidal depth profile. Optimum frequency and amplitude were empirically calculated for minimal harmonic excitation in the force sensor, so that the motion remains quasi-static. Two different palpation primitives were employed: 1) Superimposed motion on position and 2) Superimposed motion on force.

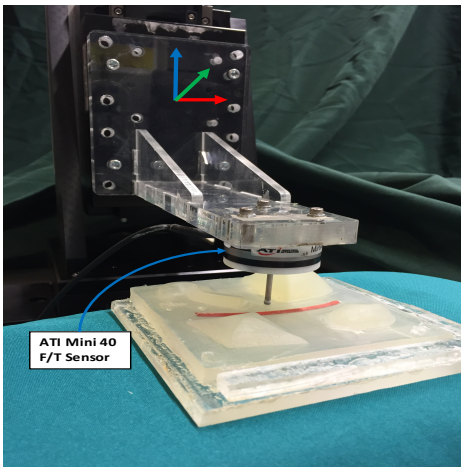


Fig. 1. Robot Setup

1) Motion Primitive I: This motion is achieved by having the desired force follow a sinusoidal profile, in the direction normal to the surface. A constant bias force in the normal direction will always be active to make sure the probe tip remains in contact with the surface at all times. The hybrid position-admittance control of the robot is given by,

$$\delta v = (I - M)\dot{\delta x} + M.\delta v_f \tag{3}$$

where, δv is the incremental velocity commanded to the robot, δx is a constant incremental step along the surface of the model and δv_f is the incremental velocity calculated based on contact force F_c and some desired bias force F_d. M is a projection matrix decomposing the forces in the normal direction and $(I - M)$ projects velocities along the surface. If the desired force F_d has a constant value, then the motion will be the same as (2). The desired force follows a sinusoidal profile which is defined through

$$F_d = F_l + F_a Sin(2\pi\omega t) \tag{4}$$

and hence

$$\delta v_f = \mathcal{G}(F_c - F_d) \tag{5}$$

where, F_l is the limiting force, F_a, ω, t are the parameters of a sin curve with F_a being the amplitude. Fig 2(a) properly depicts the motion generated by (3) and (4).

2) Motion Primitive II: This motion is achieved by making the position follow a sinusoidal profile, in the direction normal to the surface. In this case motion due to positional increment is allowed in the direction normal to the surface. Thus, the velocity profile becomes

$$\begin{aligned} \delta v &= \dot{\delta x^*} + M\delta v_f, \\ &= \delta x + \hat{n} z_a Sin(2\pi\omega t) + M\delta v_f \end{aligned}$$

where $\delta x*$ represents the new incremental step calculated based on actual incremental step δx along the surface and de-

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sired incremental step $\hat{n}z_a Sin(2\pi\omega t)$ in the normal direction $\hat{n}$. Figure 2 depicts the motion generated by this primitive.

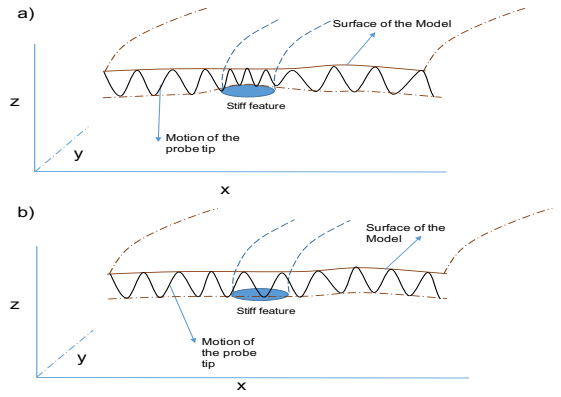


Fig. 2. a) Motion Primitive I b) Motion Primitive II

III. DIRECT GAUSSIAN PROCESS MODELLING

A Gaussian Process [18], [19] defines a probabilistic model over functions, it is commonly defined according to

$$f \sim \mathcal{GP}(\mu(x), k(x, x')), \tag{6}$$

and is completely described by its mean $\mu(x)$ and a covariance function $k(x, x')$. Given n observations, $\bar{y} = \{y_1, \ldots, y_n\}$, each observation from data set $\bar{y}$ can be assumed to be sampled from $n-variate$ Gaussian distribution. For simplicity, prior mean of a GP is usually assumed to be zero. In our model, the covariance function used is a squared exponential kernel, because it is flexible and physically intuitive:

$$k(x, x') = \sigma_f \exp\left[\frac{-\|x - x'\|^2}{2l^2}\right] + \sigma_n^2 \delta(x, x'), \tag{7}$$

where σ_f is the maximum allowable covariance and l is length scale. Gaussian white noise with variance σ_n is added to the covariance through the Kronecker delta function $\delta(x, x')$.

Since the observations are assumed to be sampled from a multivariate gaussian distribution, the predictive mean and variance at a query point x_* are expressed as

$$y_*|\bar{y} \sim \mathcal{N}(k_* K^{-1}\bar{y}, k_{**} - k_* K^{-1} k_*^T),$$

or, equivalently, the mean and variance of $f(x_*)$ are

$$\mu_f(x_*|\bar{y}) = k_* K^{-1}\bar{y}, \tag{8}$$
$$V_f(x_*|\bar{y}) = k_{**} - k_* K^{-1} k_*^T \tag{9}$$

using the terms

$$K = \begin{bmatrix} k(x_1, x_1) & \ldots & k(x_1, x_n) \\ \vdots & \ddots & \vdots \\ k(x_n, x_1) & \ldots & k(x_n, x_n) \end{bmatrix}, \quad k_{**} = k(x_*, x_*),$$
$$k_* = [k(x_*, x_1) \ldots k(x_*, x_n)].$$

Now, using the above formulation, we define a GP model over force field $\tau(x) \sim \mathcal{GP}(\mu_\tau(x), k_\tau(x, x'))$ where each point x corresponds to a palpation point $p = (p_x, p_y, p_z) \in \mathbb{R}^3$ along the palpation trajectory described in Section II-C.

A. Stiffness Estimation

To estimate stiffness, κ we need predicted forces at two different level. For simplicity, we are calculating the stiffness along the z-direction. We predict the forces on a point p_c close to the surface, which can be acquired while palpation. Forces are also estimated at a second point set p_{cd} which is d units vertically below p_c, thus resulting in the following mean and variance of stiffness

$$\mu_\kappa(p_c) = \frac{\mu_\tau(p_{cd}) - \mu_\tau(p_c)}{d} \tag{10}$$
$$V_\kappa(p_c) = \sqrt{\frac{V_\tau^2(p_c) + V_\tau^2(p_{cd})}{d^2}}. \tag{11}$$

B. Surface Geometry

Once we have the estimation of the stiffness, we can calculate the mean and variance of the surface geometry according to

$$\mu_s(p_{cd}) = p_{cd} + \frac{\mu_\tau(p_{cd})}{\kappa}.$$

Due to the non-linear relationship between stiffness and surface, the resulting covariance can be approximated [20] according to

$$V_s(p_{cd}) = J\Sigma J^T,$$

where

$$J = \left[\frac{1}{\kappa} \quad -\frac{\mu_\tau(p_{cd})}{\kappa^2}\right], \Sigma = \begin{bmatrix} V_\tau(p_{cd}) & \\ & V_\kappa(p_{cd}) \end{bmatrix}.$$

Algorithm 1 GP Batch Process

Input: Palpated positions and forces, I = $\{\vec{p_i}, \tau_i\}$
Output: $\mathcal{N}(\mu_\kappa, V_\kappa), \mathcal{N}(\mu_s, V_s)$
$I_s(p_s, \tau) = SubSample(I)$
$X = p_s$, $Y = \tau$
$GPTrain(X, Y)$
$\mu_\tau(p_c), V_\tau(p_c) = GPEstimate(p_c)$
$\mu_\tau(p_{cd}), V_\tau(p_{cd}) = GPEstimate(p_{cd})$
$\mathcal{N}(\mu_\kappa, V_\kappa) = \mathcal{N}(\frac{\mu_\tau(p_{cd}) - \mu_\tau(p_c)}{d}, \sqrt{\frac{V_\tau^2(p_c) + V_\tau^2(p_{cd})}{d^2}})$
$\mathcal{N}(\mu_s, V_s) = p_{cd} + \frac{\mu_\tau(p_{cd})}{\kappa}, J\Sigma J^T$

IV. LATENT GAUSSIAN PROCESS ESTIMATION

Instead of imposing a model directly on the force field measurements, we also consider the construction of models on the underlying physical properties of the palpated object. In our case these include the physical geometry and the surface stiffness. A key advantage of such an alternative approach is that prior information about these properties and about their relationship to force measurements could be incorporated as part of the estimation process. This is in contrast to first estimating an unconstrained force field and then deriving the surface and stiffness from the force model.

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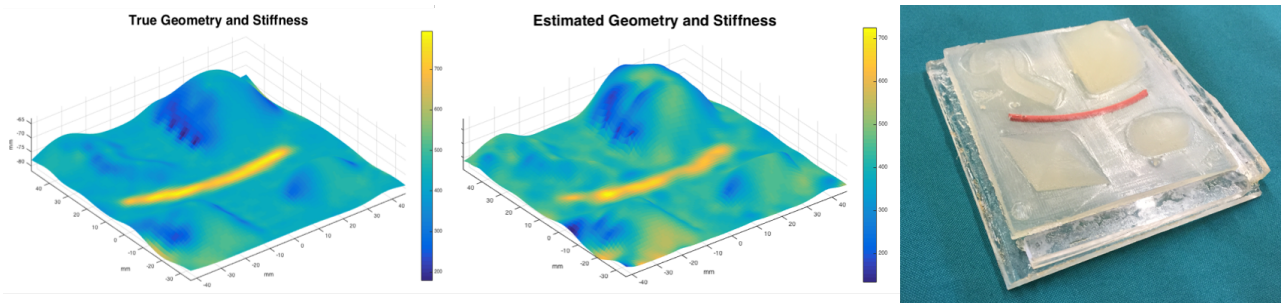


Fig. 3. True Surface and geometry

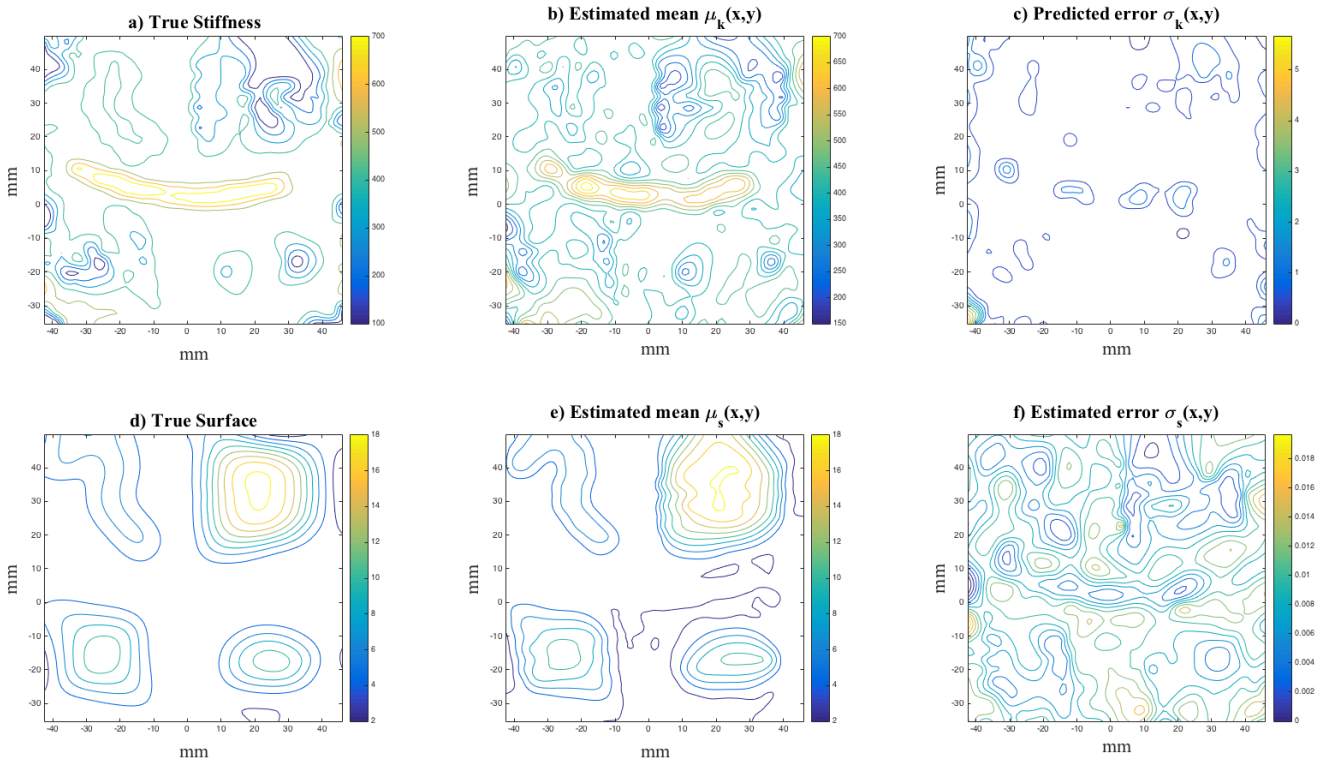


Fig. 4. Direct GP estimation a) Contour plot of True stiffness; b) Contour plot of estimated mean of stiffness; c) Contour Plot of estimated error in stiffness; d) Contour plot of True surface height; e) Contour Plot of estimated mean of the surface geometry; f) Contour plot of estimated error in surface geometry

Our approach is therefore to place a GP model over the surface $s(x) \sim GP(\mu_s(x), k_s(x,x'))$ and over the stiffness $\kappa(x) \sim GP(\mu_\kappa(x), k_\kappa(x,x'))$. Existing knowledge about the stiffness spatial correlation (e.g. size of abnormal tissue) and about the variability and roughness of the surface can be captured through proper choice of GP covariance parameters. For simplicity, we will again assume that the object geometry can be represented as a graph of the height p_z over the 2-D (p_x, p_y) domain. Similarly, we will be concerned with estimating the stiffness near the object surface which motivates representing stiffness using a graph over the (p_x, p_y) domain.

The key challenge is that there are no direct measurements of stiffness $\kappa(x)$ or surface $s(x)$ at locations $x = (p_x, p_y)$. We propose to treat the output data of these GPs as *hidden variables* that will be optimally estimated using the *induced measurements* of the forces obtained at 3-D locations p. We first introduce a general approach to latent GP estimation and then demonstrate its application to the palpation scenario.

A. Nonlinear Regression using Latent GP estimation

Consider the problem of estimating a vector-valued function $f : \mathbb{R}^{n_x} \to \mathbb{R}^{n_y}$, defined according to

$$y = f(x) \equiv \begin{bmatrix} f_1(x) \\ \vdots \\ f_n(x) \end{bmatrix}$$

with inputs $x \in \mathbb{R}^{n_x}$ and outputs $y \in \mathbb{R}^{n_y}$.

Assume that y cannot be directly observed and instead we have access to noisy observations $z \in \mathbb{R}^{n_z}$ such that

$$z(x) = h(x, f(x)) + \epsilon(x),$$

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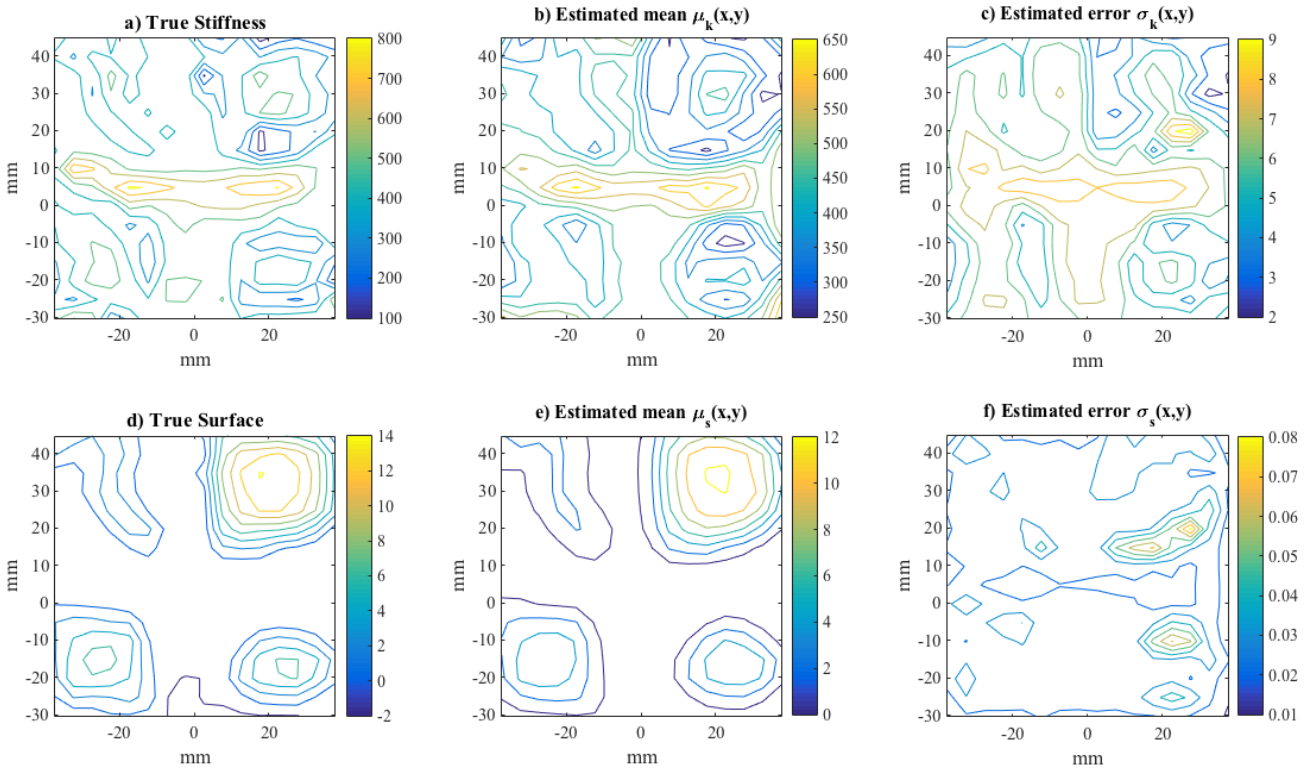


Fig. 5. Latent GP estimation a) Contour plot of True stiffness; b) Contour plot of estimated mean of stiffness; c) Contour Plot of estimated error in stiffness; d) Contour plot of True surface height; e) Contour Plot of estimated mean of the surface geometry; f) Contour plot of estimated error in surface geometry

where $h(x,y)$ is a known nonlinear function with additive Gaussian noise $\epsilon(x) \sim \mathcal{N}(0, \Sigma_\epsilon(x))$. The goal is to optimally estimate f using a set of M_z collected observation pairs $D_z = \{(x_j, z_j)\}_{j=1}^{M_z}$.

Although $f(x)$ is unknown, we assume that we have some knowledge about the spatial correlation between its inputs and outputs which motivates its representation as a Gaussian Process, i.e. $f(x) \sim GP(\mu(x), k(x,x'))$. Note that we can assume that all outputs f_i and f_j for all $i \neq j$ are uncorrelated in which case $k(x,x') \in \mathbb{R}^{n_y \times n_y}$ is diagonal or that they are coupled [21] in which case $k(x,x')$ will have off-diagonal elements. Normally the posterior of such a GP would be constructed given data pairs $D_y = \{x_j, y_j\}_{j=1}^{M_y}$ which in our case are unavailable, or latent. Our approach is to select the points x_j in D_y using some rule (e.g. they could be a subset of those in D_z or otherwise could be placed on a uniform grid over the domain of interest) and to regard the outputs y_j in D_y as unknown variables to be optimally estimated. In addition, the GP hyperparameters θ_{hyper} (e.g. the maximum covariance and length-scale using in standard kernels) could also be treated as unknowns. We thus form the vector of latent variables $\theta = (\bar{y}, \theta_{\text{hyper}})$ where $\bar{y} = (y_1, \ldots, y_{M_y})$ and assume that a prior is available as

$$\theta \sim \mathcal{N}(\cdot|\theta_0, P_0),$$

with a known mean θ_0 and covariance matrix P_0.

The most likely GP model of f can now be estimated using a maximum-likelihood approach by optimizing

$$\begin{aligned} &\min_\theta -\log p(f|D_z, D_y) \\ &= \min_\theta -\log p(\bar{z}|f, \bar{x}, \bar{y}) - \log p(f|\bar{x}, \bar{y}) \\ &= \min_\theta \sum_{j=1}^{M_z} \|h(x_j, f(x_j;\theta)) - z_j\|^2_{\Sigma_\epsilon^{-1}(x_j)} + \|\theta - \theta_0\|^2_{P_0^{-1}}, \end{aligned} \tag{12}$$

where the GP form of f is given by

$$f(x_j;\theta) = \bar{k}(x_j)^T K^{-1} \bar{y},$$

where

$$\bar{k}(x) = \begin{bmatrix} k(x,x_1) \\ \vdots \\ k(x,x_{M_y}) \end{bmatrix}, K = \begin{bmatrix} k(x_1,x_1) & \cdots & k(x_1,x_{M_y}) \\ \vdots & \ddots & \vdots \\ k(x_N,x_1) & & k(x_N,x_N) \end{bmatrix}.$$

The formulation (12) corresponds to a nonlinear least-squares problem in view of the residual vector $r = (r_1, \cdots, r_{M_z})$ where each residual is defined by

$$r_j(x;\theta) = h(x, f(x;\theta)) - z_j, \quad j = 1, \ldots, M_z.$$

The gradient of each residual can be readily computed as

$$\nabla_\theta r_j(x,\theta) = K^{-1}\bar{k}(x) \nabla_y h(x, f(x)).$$

A desirable property of the gradient $\nabla r(\theta)$ is that $\nabla r(\theta) \nabla r(\theta)^T$ is full rank, which requires that $M_z \geq M_y n_y$. Note that all quantities $K^{-1}\bar{k}(x_j)$ are precomputed and

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cached before a least-squares solver such as Levenberg-Marquardt is applied.

The resulting error covariance of the estimate θ^* can be approximated as

$$P \approx \frac{1}{M_z - M_y n_y} \|r(\theta)\|^2 (\nabla r(\theta) \nabla r(\theta)^T)^{-1}.$$

in case of uninformative priors (with $P_0 = \infty$). This expression can be easily extended to the case when a prior is employed.

B. Application to palpation-based model construction

We next apply the proposed latent GP approach to the palpation problem. The measurements $z \in \mathbb{R}_{\geq 0}$ denote the norm of a force recorded by the robot at a commanded end-effector point $p = (p_x, p_y, p_z) \in \mathbb{R}^3$. Our model has two hidden outputs $y = (\kappa, s)$ where κ denotes the surface stiffness and s denotes the surface height over a given point $x = (p_x, p_y)$. The function h then defines the physical relationship between force, depth, and stiffness according to

$$h(x, y) = \kappa(s - p_z(x)),$$

where the function $p_z(x)$ simply looks up the recorded height at given recorded location $x = (p_x, p_y)$. We assume that the force measurements have variance $\Sigma_\epsilon = \sigma_f^2$ with $\sigma_f = 0.01$ N.

We model the two outputs as independent GPs in which case we have a diagonal kernel covariance

$$k(x, x') = \begin{bmatrix} k_\kappa(x, x') & \\ & k_s(x, x') \end{bmatrix},$$

where k_κ defines the GP covariance for the stiffness and k_s the one for the surface. We employ a standard squared exponential kernel for both with different parameters.

The gradient of the force required for the solution of (12) is simply

$$\nabla_y h(x, y) = \begin{bmatrix} s - p_z(x) \\ \kappa \end{bmatrix}.$$

V. ADAPTIVE SAMPLING

The key advantage of using GPs to estimate object properties, including stiffness and geometry, lies in the availability of uncertainty confidence intervals. By predicting both the mean and the variance one can adaptively guide palpation towards informative regions of interest. This is accomplished through local continuous tool-tip path optimization aiming to: 1) reduce the overall model uncertainty, or 2) locate abnormally stiff features as quickly as possible.

The optimized adaptive sampling is formulated as a stochastic optimization over trajectories parametrized using a finite dimensional vector ξ. The parameter encodes a sequence of n_s arcs in 3-D space and is defined as

$$\xi = (v_1, \omega_1, d_1, \ldots, v_{n_s}, \omega_{n_s}, d_{n_s}),$$

where v_i denotes the horizontal forward velocity, ω_i the horizontal angular velocity, d_i the rate of change of depth. The motion is generated by assuming that the tip moved with constant velocity (v_i, ω_i, d_i) during time inverval $[t_{i-1}, t_i]$, for each $i = 1, \ldots, n_s$ with $t_0 = 0$ and t_{n_s} provided as a fixed parameter.

Under this parametrization at time $t \in [t_i, t_{i+1}]$ the position of the tip $p(t)$ and its heading angle $\theta(t)$ are

$$p_x(t) = \begin{cases} p_x(t_i) + \frac{v_{i+1}}{\omega_{i+1}}(\sin\theta_{i+\Delta} - \sin\theta_i), & \text{if } \omega_{i+1} \neq 0 \\ p_x(t_i) + v_{i+1}\Delta t_i \cos\theta_i, & \text{otherwise} \end{cases}$$

$$p_y(t) = \begin{cases} p_y(t_i) + \frac{v_{i+1}}{\omega_{i+1}}(\cos\theta_i - \cos\theta_{i+\Delta}), & \text{if } \omega_{i+1} \neq 0 \\ p_y(t_i) + v_{i+1}\Delta t_i \cos\theta_i, & \text{otherwise} \end{cases}$$

$$p_z(t) = p_z(t_i) + \Delta t_i d_{i+1}$$

$$\theta(t) = \theta(t_i) + \Delta t_i \omega_{i+1},$$

where $\Delta t_i = t - t_i$, $\theta_i = \theta(t_i)$, and $\theta_{i+\Delta} = \theta_i + \Delta t_i \omega_{i+1}$.

We consider two different cost functions defined for given candidate paths $p(t)$: one minimizing the variance of stiffness, and thus seeking to improve the overall model quality, and the other seeking maximum stiffness. The costs are defined, respectively, by

$$J(\xi) = \begin{cases} \int_{t_0}^{t_f} -V_\kappa(p(t))dt, & \text{: variance,} \\ \int_{t_0}^{t_f} -\mu_\kappa(p(t)) - \beta\sqrt{V_\kappa(p(t))}dt & \text{: UCB,} \end{cases}$$

where β is a user-defined constant determining the upper confidence bound (UCB) on stiffness estimate and is used to balance between visiting known stiff regions and exploring uncertain but potentially stiffer regions. We set $\beta = 1.96$ corresponding to 95% confidence in the stiffness estimate. Both costs are computed by approximating the integrals using a finite number of points p_i along $p(t)$ and predicting the mean $\mu_\kappa(p_i)$ and variance $V_\kappa(p_i)$ from the current GP models.

The actual optimization of $J(\xi)$ is performed using the cross-entropy method [22], [23] which is a global optimization method that places a Gaussian density over ξ and iteratively updates this density until it has concentrated around the optimal ξ^* minimizing the cost function. This method was chosen since it can handle non-smooth and highly irregular cost landscapes such as those that occur during non-parametric estimation.

VI. RESULTS AND DISCUSSION

We applied the approach discussed in Section III to a palpation set which consists of 77000 palpation points and force pairs. For verification purposes, we initially estimated the true stiffness and geometry using all the palpation points. However, a subset of n = 2500 points were used for training and a different set of m = 1000 points were used for prediction. For display purposes we interpolated the estimated stiffness and geometry over a regular grid with 1mm cell size. For this experiment, we used a length scale of 5mm and optimized just the variance parameter,σ_f, by maximizing the negative log-likelihood, as described in [18]. The resultant value of σ_f was 0.86, which complies with the observations, because the maximum force applied on the object was around 1.6N and it lies in the 95% confidence level of the maximum variance.We approximately modeled the noise parameter, $\sigma_n = 0.01N$, by analyzing the noise observed in force measurements measured when various

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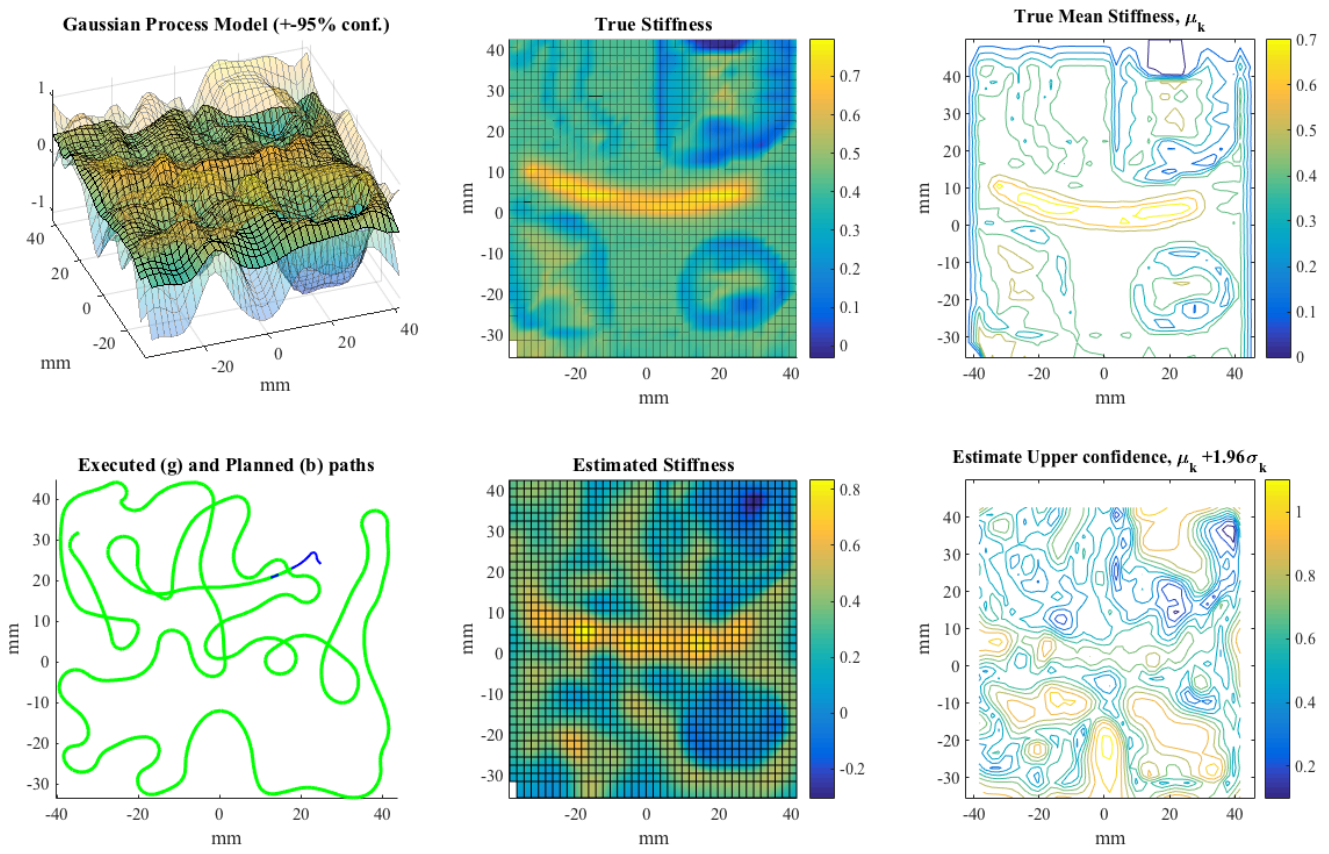


Fig. 6. Adaptive sampling: a) GP Model over Stiffness; b) True Stiffness; c) Contour Plot of the true stiffness; d) Executed trajectory path; e) Estimated stiffness f) Estimated upper confidence level of stiffness

static loads were applied on the sensor. Fig 4 compares the resulting estimation to the ground truth estimates from 77000 samples, in Fig 3. Stiffness error for this particular set of samples was 120.4 mN/mm and the root mean squared error of the surface geometry was 2.6161mm. This accuracy in stiffness/surface estimation is sufficient for a surgeon to detect stiff features or abnormal growth of tissue.

We also tested our approach against varying sizes of input samples. The variance in the force field decreases as the number of training samples were increased. This makes sense because, it is expected to see a low uncertainty in the prediction of an unknown observation, when the prediction is based on large number of neighboring known observations. RMS error in surface geometry however decreases very slowly as the number of training samples are increased. The reason for this behavior is because of the way we are calculating the surface geometry. The error in surface estimation is not only propagated from the force uncertainty, but also from stiffness uncertainty. An alternative way of calculating the surface geometry, using the force field, is to do a linear search in vertical direction, at each grid point, and look for the distribution where the mean is zero and has the minimum variance. This will give an estimate of the surface, with a much higher confidence than the one computed using the predictive distribution of stiffness.

Adaptive sampling was tested by initializing at a random location. It was interesting to observe, in the executed path shown in Fig 6d, that the simulated trajectory constantly follows a spiral motion at stiff regions or unexplored regions, which is identical to what a person would intuitively do, to characterize the surface properties. This shows that the trajectory, simulating the motion of the probe, adaptively changes the trajectory parameters to minimize the uncertainty at unexplored and stiff regions. Fig 6e reveals the underlying stiff feature based on the executed trajectory in Fig 6d. However, a significant noise is observed in the estimated stiffness field, and it is likely due to incorrect force interpolation at the current location $p(t)$. We did multiple restarts

We have also applied the approach, described in IV, to the palpation problem using $M_z = 2000$ data points by selecting the locations x_j in the latent GP dataset D_y to be on a regular grid with 5mm cell size resulting in $M_y = 256$ points. For this experiment we employed very wide priors specifying that surface mean is at 15 mm with standard deviation of 20 mm and a stiffness at 500 N/m with standard deviation of 1000 N/m were used. Figure 5 compares the resulting estimated function to the ground truth estimates from 77000 points.

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VII. CONCLUSIONS AND FUTURE WORK

This paper explored different ways to employ Gaussian Processes for concurrent estimation of surface geometry and stiffness, based on continuous palpation primitives. One is specification of GP model over force field. Estimated distribution of force is then used to predict the probabilistic mean of stiffness and geometry, along with variance which is estimated by error propagation. For the second approach, individual GP models were specified over geometry and stiffness. This approach treats the output as hidden variables, which are optimally measured using induced force measurements. Correlation between geometry and stiffness is captured by having proper choice of GP covariance parameters.

These approaches were validated on data collected by continuous motion primitives, based on some model assumptions. We also provided a means to adaptively sample the robot trajectory, using stochastic optimization on a bayesian cost, for collecting optimal number of samples, to have maximum information gain. This helps reduce the palpation time, as the velocity changes adaptively based on uncertainty at the feature of interest.

In the future, we plan to repeat our experiments on *ex-vivo* organs and will extend our results to account for gross motions of the tissue. We will also explore path optimization techniques for faster task execution, which will include hybrid strategies of combining coarse fast palpation to identify regions with slow accurate palpations to get better values or delineate edges. We will also investigate Multi-output Gaussian Processes as an alternative for concurrent surface and stiffness estimation.

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